Search in Whose Language? Separating the Query-Side Gap from the Evidence-Side Gap for Search Agents in Indic Languages

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Abstract

When a user asks an LLM search agent a question in Hindi or Telugu, the answer is less often right than when the same question is asked in English. Two different things can cause that, and they call for different fixes: the agent may search badly, or in a language where the evidence is not (the *query-side gap*), or the evidence may not exist in that language (the *evidence-side gap*). We separate them. We hold the question fixed with human-parallel Hindi/English and Telugu/English questions whose answer is verifiably in both Wikipedias (287 Hindi, 127 Telugu items), hold the evidence fixed with a frozen pool of 1.9 million Wikipedia passages served English-only, Indic-only, or both, log the script of every search query the agent writes under a free policy, and then force the query language. Four models (three Gemini, one open-weight Qwen3) run the 19-cell grid under a cross-lingual dense retriever and again under BM25. Left free, agents write 18–57% of their queries in English and do not switch when a pool returns no Indic text. The cheapest models lose 9 points on the Hindi question with the dense retriever; the two stronger Gemini models lose 2–3 points, inside the noise. Where the gap exists, forcing the query language recovers at most a few points and the served pool makes no difference, because a Hindi query already finds the English passage; with the gold passage in the prompt the Hindi question is still answered 4–6 points worse, so what remains is reading. Under BM25 the same items tell the opposite story: the Hindi question costs 19–20 points, dual-language queries recover 9–14 of them, and Hindi-only queries on a Hindi-only pool collapse. We measure that the Hindi and Telugu Wikipedias state the answer for only 29% and 24% of the questions, and a coverage-weighted decomposition shows the consequence: with keyword search a real Hindi user gets 45 against 71 for an English user and searching in English lifts her to 63; with cross-lingual search she gets 67 against 76 and the query language no longer matters. The query-side gap is a property of the search engine; the residue belongs to the model. Six directions were pre-registered; three held, two failed, one split.

1 Introduction

Search agents, a language model that writes queries, reads what comes back, and answers, are now how hundreds of millions of people in India reach the web through chat products. When the question is in Hindi, Telugu, or Bengali rather than English, the answer is wrong more often. That much is established for browsing and deep-research agents across many languages (Hofman et al., 2025; Caciolai et al., 2026; Lu et al., 2026; Aji et al., 2026). What is not established is *why*, and the why decides the fix.

Two causes produce the same drop. The first is the agent. It may write a worse query in Hindi than in English, or it may write its query in a language where the evidence is not:

an English query for a fact that only the Hindi Wikipedia records, or a Hindi query for a fact that only the English Wikipedia records. We call this the **query-side gap**. Prompting and training can fix it. The second is the web. Hindi Wikipedia has 2.4% of the articles of English Wikipedia, Telugu 1.8%, and Hindi is 0.2% of Common Crawl pages. If the fact is not written in the user's language, no query will find it in that language, and the only remedies are cross-lingual search or writing the content. We call this the **evidence-side gap**. A residual remains when the right passage is in front of the model and it still answers worse in Hindi than in English; we call that the **integration gap**.

Nobody has separated these for a search agent. The closest study translates an English deep-research corpus into twelve languages and measures how much oracle retrieval recovers, but the question is always English, the agent's own query language is never observed or manipulated, and no Indic language is included (Lu et al., 2026). A SIGIR 2026 study evaluates two web-search models on native Marathi queries and attributes failures to integration "rather than missing sources" by inspection, with one language and no manipulation (Kale, 2026). Single-shot retrieval work has named the confound, that an apparent preference for English evidence is partly an artefact of where gold evidence lives (Park et al., 2026), and has shown, with fixed context and no retrieval, that models prefer the query's language when a fact exists only in one Wikipedia edition, with Bengali the worst case (Ki et al., 2026). English-only failure taxonomies split agent failures into "never found" and "found but misused" (Liu et al., 2026b; Liang et al., 2026). None of these logs what language an agent searches in when a user asks in an Indic language, forces that language, and serves or withholds the evidence in each language on parallel questions.

We do that (Figure 1). Our instrument has three parts. **Parallel questions**: human-translated Hindi/English and Telugu/English question pairs from IndicGenBench XorQA-In (Singh et al., 2024) and natively parallel Hindi/English Wikipedia paragraphs from MLQA (Lewis et al., 2019), kept only when the Indic Wikipedia article verifiably states the answer, so that the evidence exists in both languages by construction. **A frozen two-language web**: the complete Hindi and Telugu Wikipedias and a sample of English Wikipedia from the same October 2026 snapshot, chunked and indexed, with the agent's single `search` tool served from the English-only pool, the Indic-only pool, or their union. **A query-language log and lever**: under a free policy we record the script of every query the agent writes; under three forced policies the agent must query in the question's language, in English, or in both. A closed-book arm measures what the model already knew, and an oracle arm with the gold passage in the prompt measures the integration gap. Because dense multilingual retrieval is cross-lingual and lexical retrieval is not, we run the grid under both a bge-m3 dense retriever and BM25: whose language the search engine speaks is itself part of the query-side gap.

The controlled grid gives conditional accuracies. To turn them into a statement about the real web we measure coverage: for 539 XorQA questions per language, whether the Hindi or Telugu Wikipedia article linked to the English source article states the answer (string match, two independent LLM reads of the article, and a manual audit). The decomposition

$$\mathrm{Acc}_L \;=\; \mathrm{Cov}_L \cdot \mathrm{Acc}(L \mid \text{evidence in } L) \;+\; (1-\mathrm{Cov}_L) \cdot \mathrm{Acc}(L \mid \text{evidence only in English}) \quad (1)$$

then says how many points of the Indic user's deficit are the web's, how many are the agent's, and how many a different query policy recovers.

Findings.

1. **Agents search in a mix of languages and do not adapt.** Left free with a Hindi question, the four models write 18–57% of their queries in English (30% for Gemini 2.5 Flash-Lite), more for Telugu (43–48%), and the share does not move when the pool returns no Indic text (Section 4.1).
2. **The gap belongs to the cheap models.** With evidence in both languages and the dense retriever, the Hindi question costs Gemini 2.5 Flash-Lite 9.4 points (95% interval 4.5 to 14.3) and Qwen3-30B-A3B 9.1 (3.8 to 14.3), both surviving correction over 134 tests; Gemini 3.1 Flash-Lite and 3.6 Flash lose 3.0 and 2.0, intervals through zero. Telugu gaps are 1–5 points and not significant (Section 4.2).

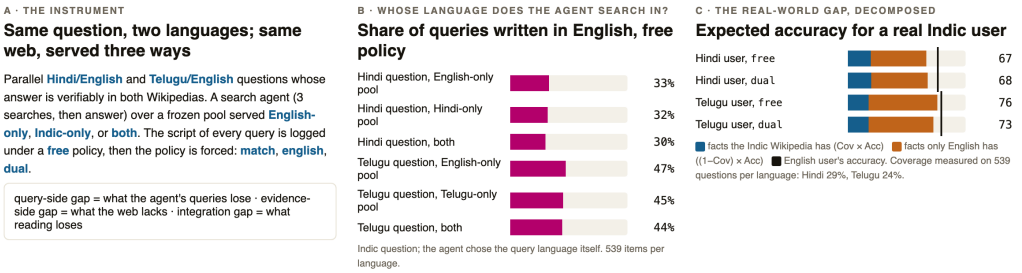


Figure 1: Study design and headline numbers (Gemini 2.5 Flash-Lite). (a) Parallel Hindi/Telugu and English questions; a frozen two-language Wikipedia pool served English-only, Indic-only, or both; the agent's query script logged under a free policy and then forced. (b) Share of queries the agent writes in English when the question is in Hindi or Telugu: 30–47%, unchanged by the pool served. (c) Expected accuracy of a real Indic user from Eq. 1: the share of facts the Indic Wikipedia holds (29% Hindi, 24% Telugu) times the both-pools accuracy, plus the share only English holds times the English-only accuracy; with the cross-lingual retriever the two terms add to 67 (Hindi) and 76 (Telugu) against 76 and 79 for an English user.

3. **Under a cross-lingual retriever the query language is not the cause.** Forcing matched, English, or dual queries changes Gemini 2.5 Flash-Lite's Hindi accuracy by +0.7, +4.5 and +1.8 points, none surviving; the English-only, Hindi-only and both pools score within one point of each other. English queries help on the English-only pool and hurt on the Hindi-only pool (interaction 11.2 points, survives), the one place policy matters (Sections 4.2, 4.3).

4. **Under BM25 the query language is everything.** The same items cost 18.9 (Gemini 2.5 Flash-Lite) and 19.5 (Qwen) Hindi points; dual queries recover 9.3 and 13.6 (both survive); Hindi-only queries on the Hindi-only pool fall to 35%; the agent abstains in a quarter to a third of episodes (Section 4.2).

5. **The web lacks the evidence, but only the keyword engine pays for it.** The Hindi and Telugu Wikipedias state the answer for 29.1% and 23.6% of the questions. Coverage-weighted, a Hindi user of Gemini 2.5 Flash-Lite with keyword search expects 45.4 against 71.4 for an English user and 62.9 if told to search in English; with cross-lingual search 66.7 against 75.8, and the policy no longer matters (Section 4.4).

6. **What remains is reading.** With the gold English passage in the prompt, the Hindi question is still answered 6.3 points worse by Gemini 2.5 Flash-Lite (survives) and 4.2 by Qwen; the cheap model reads the Hindi passage worse than the English one (72.8 vs 80.5); the stronger Gemini models read as well in Hindi as in English (Section 4.5).

Pre-registration. Six directional predictions (D1–D6) were written into the project log before any grid call, with their falsification conditions. D3, D5 and the first half of D4 held; D1 and the second half of D4 failed; D2 and D6 split by model. Each is scored where it is tested in Section 4.

Responsible framing. Only public Wikipedia text (CC BY-SA) and public benchmark questions are used; the pools regenerate from the public snapshot and our scripts. Models are queried through their public APIs or an open-weight checkpoint on our own GPU. No deployed search product is probed.

2 What was known and what is new

We want to be plain about what is already established. That LLM agents do worse on non-English tasks is known: GAIA in eleven languages scores 47.4% in English against 35.8% elsewhere, Hindi included (Hofman et al., 2025); GAIA2 in ten languages finds Hindi the weakest at every model size (Caciolai et al., 2026); natively authored browsing questions in thirteen languages differ by language (Aji et al., 2026). That retrievers and generators prefer English and query-language passages is known (Park and Lee, 2025; Wang et al., 2026; Ki et al., 2025). That the apparent English preference is partly an artefact of where gold evidence lives is known for single-shot RAG (Park et al., 2026). That models write tool arguments in the wrong language is known, Hindi included (Luo et al., 2026). That parametric knowledge of India-centric facts is weaker in Indic scripts is known (Rohera et al., 2025; Mazumder et al., 2026; Hupkes and Bogoychev, 2025; Ki et al., 2026). That agent failures split into retrieval and utilisation is known for English (Liu et al., 2026b; Liang et al., 2026; Liu et al., 2026a).

Closest neighbour	What it established / what it does not cover
XBCP (Lu et al., 2026)	Deep-research agents on BrowseComp-Plus with the corpus translated into 12 languages; oracle decomposition into retrieval vs integration. Question always English; query language never logged or forced; no Indic language; translated rather than native evidence.
MarathiWeb (Kale, 2026)	Two web-search LLMs on native Marathi queries; failures attributed to integration by inspection. One language; evidence availability observed, not manipulated; no parallel English arm.
MAST Indic track (Thakur et al., 2026)	Nine Indic query languages over the English BrowseComp-Plus corpus. English-only corpus, so the evidence-side gap is zero by construction; a systems shared task.
DELTA (Park et al., 2026)	Names the gold-availability prior and regresses it out of single-shot RAG. No agent, no Indic, statistical rather than experimental control.
Where's Waldo (Ki et al., 2026)	Query-language preference with fixed paired context when a fact exists in one edition only; Bengali worst. No retrieval, no agent, no search-query language.
Diagnosing search failures (Liu et al., 2026b)	Retrieval gap vs utilisation gap with oracle evidence. English only; the retrieval gap is not split into query formulation vs corpus coverage.
Cross-Lingual Cost (Amiraz et al., 2025); XRAG (Liu et al., 2025); tRAG/CrossRAG (Ranaldi et al., 2025); CORAL (Lee et al., 2026b)	Query-language vs document-language effects in RAG; one controlled Google ablation; an agent that picks the corpus language. Non-agentic or end-accuracy only; no Indic; no decomposition.

Table 1: Novelty delineation. Full related work in Section 6.

What is new here is (i) the agent's search-query language as a measured mediator and then a manipulated factor, (ii) parallel Indic/English questions over a frozen evidence pool in which the language of the gold evidence is served or withheld, (iii) the retriever (cross-lingual dense vs monolingual lexical) as a factor in the query-side gap, and (iv) a coverage term measured on the live Indic Wikipedias that weights the controlled accuracies into a real-world decomposition.

3 Method

3.1 Questions

XorQA-In (Singh et al., 2024) provides 539 test questions per language, human-translated from the English TyDi XOR questions (Asai et al., 2020), each with the English Wikipedia paragraph that answers it and the English answer. For each item we located the Hindi

or Telugu counterpart of the English source article through Wikidata sitelinks (916 of 918 titles resolved) in the same snapshot and checked whether that article states the answer: by answer-string match in either script, then by two independent LLM reads of the full article (gemini-2.5-flash-lite, then gemini-3.6-flash) that had to quote a sentence found verbatim in the article, with a 50-item manual audit (Appendix B). An item enters the controlled grid only when both reads agree and the quoted or matched sentence exists: 157 Hindi and 127 Telugu items. The Telugu translated answers in the release are unusable for 237 items (the string "అని", "that"), so Telugu answers are scored against the English gold and by an LLM judge. **MLQA** (Lewis et al., 2019) provides 4,918 Hindi questions whose English counterparts share ids and whose Hindi and English paragraphs are naturally parallel Wikipedia text; many MLQA questions presuppose their paragraph ("What did Keane often request?"), so we kept only questions an LLM filter judged self-contained (176 of 450 checked) and sampled 130. Both gold paragraphs are injected into the pools. **TyDi XOR** Telugu validation questions (120 sampled from 291 answerable), natively written rather than translated, form a robustness subset for the translation-artefact concern (Kim et al., 2026); they have no English twin and are run in Indic-question cells only. Every item carries its English and Indic gold answers; the full bank with the coverage flags is frozen with SHA-256 hashes.

3.2 The frozen two-language web

All pools come from one snapshot, `wikipedia-monthly` 20261001 (Kamali, 2026): the complete Hindi Wikipedia (169,949 articles, 353,075 chunks), the complete Telugu Wikipedia (145,514 articles, 518,524 chunks), and a 1.8% sample of English Wikipedia (129,646 articles, 1,058,175 chunks) to which we add the full English article behind every item (681 articles, 7,310 chunks) and 666 deduplicated gold paragraphs. Articles are split into chunks of about 900 characters with the title prefixed. The agent's one tool, `search(query)`, returns the top five chunks from the pool being served: English-only, Indic-only, or their union. We run two retrievers. BM25 (Robertson and Zaragoza, 2009; Lù, 2024) over Unicode word tokens with no stemming is the lexical instrument; bge-m3 dense embeddings (Chen et al., 2024) with exact inner-product search are the semantic one. A sanity check with the question itself as the query put the gold chunk in the top five for 50% of English questions under BM25 but only 8% of Hindi and 7% of Telugu ones (transliteration variants such as ओरेगन/ऑरेगॉन and Telugu case suffixes defeat exact tokens); bge-m3 gave 70% for both Indic languages and retrieved Hindi gold for English queries 68% of the time. The dense retriever is therefore the main-grid instrument and BM25 the comparison; the difference between them is part of what we measure.

3.3 The agent and the policies

A fixed ReAct loop (Yao et al., 2022) over native tool calling: a system prompt, the question, at most three `search` calls, then a short final answer in the question's language; temperature 0; minimal reasoning effort. The only text that differs between policy cells is one sentence: none (**free**), "write every query in [the question's language]" (**match**), "write every query in English" (**english**), or "at every step search once in each language" (**dual**). For every search call we log the query, its dominant Unicode script (Devanagari, Telugu, Latin, or mixed), the languages of the five hits, and whether a gold chunk was among them. Verbatim prompts are in Appendix A.

3.4 Factors, arms, and models

Question language $Q \in \{\text{en}, L\}$; policy $P \in \{\text{free, match, english, dual}\}$; evidence pool $E \in \{L\text{-only, both}\}$. The core grid is $Q{=}L{\times}4P{\times}3E$ plus $Q{=}\text{en}{\times}\{\text{free, match}\}{\times}3E$ plus $Q{=}\text{en} \times \text{dual} \times L\text{-only}$: 19 cells, every factorial item, both languages, under the dense retriever; the same 19 cells under BM25 for the cheapest model and the eight headline cells for the others. Arms: **closed-book** (no tool, $Q \in \{\text{en}, L\}$) for parametric leakage; **oracle** (gold passage in the prompt, $Q \in \{\text{en}, L\}$) for the integration gap; **none** (the pool with the item's gold removed, 50 items per language) for the abstention floor; a **repeat battery** (30 items $\times$ 5 identical runs) for the noise floor; and the **native-Telugu** subset.

Models: gemini-2.5-flash-lite (full grid and all arms), gemini-3.1-flash-lite (120 items per language), gemini-3.6-flash (headline cells, 60 items per language), all through OpenRouter, and Qwen3-30B-A3B-Instruct-2507 served with vLLM on a rented GPU as the open-weight family. Other closed families were blocked by a workspace guardrail on the API account at the time of the study (Section 7).

3.5 Scoring and statistics

An answer is correct if it matches a gold alias in either script after normalisation (digit systems unified, 40-character containment slack), otherwise a judge model not in the grid (gemini-2.5-flash) labels it CORRECT, WRONG, or ABSTAIN from the English and original question and all gold aliases; 40 judged items are hand-audited (Appendix C). The unit is the item. Every contrast is a paired within-item difference with a sign-flip permutation test (20,000 draws) and an item-bootstrap 95% interval (10,000 draws); the policy × evidence interaction is tested by label permutation (5,000); all reported p-values form one Benjamini–Hochberg family (Benjamini and Hochberg, 1995) and only survivors are set in bold. Every number traces to one analysis script.

4 Results

Table 2 lists every paired contrast with its interval, permutation p-value and Benjamini–Hochberg status; Appendix D lists every cell. All numbers are accuracies in percentage points on the factorial items (287 Hindi, 127 Telugu). The repeat battery (30 items × 5 identical runs, Indic question, free policy, both pools) changed the verdict on 1 of 30 Hindi items and 0 of 30 Telugu items, so cell-level differences below about two points are within the noise floor and are not interpreted.

4.1 In what language do agents search?

Under the free policy, with the question in Hindi, Gemini 2.5 Flash-Lite writes 30–33% of its search queries in English, Gemini 3.1 Flash-Lite 31–38%, Gemini 3.6 Flash 45–57%, and Qwen3-30B-A3B 18–19% (Figure 2). With the question in Telugu every model searches in English more often: 44–48% for the three Gemini models and 43–46% for Qwen. Mixed-script queries are rare (0–3%, 15% for Gemini 3.6 Flash on the English-only pool). The shares barely move with the pool that is served: the agents do not notice that an English-only pool returned no Hindi text and switch. Pre-registered direction D1 predicted a majority of English queries; that holds for one model and one pool (Gemini 3.6 Flash, Hindi question, English-only pool, 57%) and fails elsewhere. The second half of D1, more English for Telugu than for Hindi, holds for every model except Gemini 3.6 Flash, where the two are equal.

4.2 The query-side gap and how much a policy recovers

With evidence present in both languages and the dense retriever, the Hindi question costs Gemini 2.5 Flash-Lite 9.4 points against the English question (66.2 vs 75.9, interval 4.5 to 14.3, survives correction) and Qwen3-30B-A3B 9.1 points (65.5 vs 74.6, survives). The two stronger Gemini models show no measurable gap: 3.0 points for 3.1 Flash-Lite (interval −2.0 to 8.0) and 2.0 points for 3.6 Flash (interval −4.0 to 8.0). For Telugu the gaps are 4.7, 1.0, −6.0 and 2.4 points, none significant. The gap is a property of the cheaper models, not of the languages.

Where the gap exists, forcing the query language does not close it. On Gemini 2.5 Flash-Lite, Hindi question, both pools: matched-language queries change accuracy by +0.7 points, dual-language queries by +1.8, English-only queries by +4.5 (interval 0.0 to 9.1, $p = 0.07$, not surviving correction). Gold passages are already in the top five for 79% of free-policy episodes and 87% of dual-policy episodes, so there was little retrieval to recover. On Qwen the dual policy recovers 4.5 of the 9.1 points ($p = 0.035$, not surviving). D2 (a query-side gap that dual queries halve) is therefore supported in direction for Qwen and not for Gemini; the measured query-side gap under cross-lingual retrieval is at most a few points.

Under BM25 the same model and items tell a different story (Figure 5, Table 2, rows marked BM25). The Hindi question now costs 18.9 points (50.3 vs 71.4, survives), the Telugu question 11.0 points, and the Indic question with the lexical retriever scores 15.7 points (Hindi) and 21.3 points (Telugu) below the same question with the dense retriever (both survive). Here the policy matters: dual queries recover 9.3 of the 18.9 Hindi points (survives), and the English-only policy on the English-only pool reaches 62.7 against 43.4 under the free policy. Gold passages reach the top five in 40% of free-policy episodes under BM25 against 79% under dense retrieval, and the agent abstains in 23–39% of BM25 episodes against 6–8% with dense retrieval. The query-side gap is large when the search engine is monolingual and nearly absent when it is cross-lingual.

4.3 Policy × evidence: when searching in English helps and when it hurts

Pre-registered direction D3 said that forcing English queries helps when the evidence is English-only and hurts when it is Indic-only. On Gemini 2.5 Flash-Lite with Hindi questions this is what happens: English queries beat matched-language queries by 6.3 points on the English-only pool (survives) and lose 4.9 points on the Indic-only pool ($p = 0.10$); the interaction is 11.2 points (interval 4.2 to 18.2, survives). The same model with Telugu questions shows no interaction (1.6 points), and neither does Gemini 3.1 Flash-Lite (3.0 and 2.0 points). The sign-flip predicted by the disagreement between query-translation studies (Amiraz et al., 2025; Ranaldi et al., 2025) is real, but only for the model and language where lexical cues still matter to a dense retriever: Hindi queries on English text retrieve the gold chunk 66% of the time and English queries on Hindi text 68%, so the policy mostly decides which pool the agent can read.

4.4 Coverage and the decomposition of the real-world gap

Of the 539 XorQA questions per language, the linked Hindi Wikipedia article exists for 67.5% and verifiably states the answer for 29.1% (interval 25.2 to 33.0); for Telugu the article exists for 52.5% and states the answer for 23.6% (20.0 to 27.1). D4's first clause holds (both below one half, Hindi above Telugu). The manual audit of 50 items (Appendix B) found the first LLM read too generous, which is why a second read was required; the coverage estimates are the two-read figures.

Table 3 and Figure 4 put the coverage into Eq. 1. With the dense retriever the evidence-side term vanishes: for every model the Indic question on the English-only pool scores within two points of the same question on both pools (Table 2, evidence-side rows), because the retriever finds the English gold for a Hindi or Telugu query. The expected accuracy of a Hindi user is therefore 66.7 (Gemini 2.5 Flash-Lite), 76.4 (3.1 Flash-Lite) and 74.0 (3.6 Flash) against 75.8, 77.0 and 76.0 for an English user; the whole remaining gap is the agent's reading, not the web. D4's second clause, that the coverage term would exceed the agent term for Telugu, fails under cross-lingual retrieval. Under BM25 it is the other way round: the Hindi user of Gemini 2.5 Flash-Lite gets 45.4 against 71.4 for the English user, the Telugu user 49.9 against 66.9, and most of that loss is the 71% of facts that only English Wikipedia holds, which a Hindi or Telugu query cannot reach lexically. Telling the agent to search in English lifts the Hindi user to 62.9 and the Telugu user to 67.1. So the advice to a product team depends on the engine: with lexical search, translate the query; with cross-lingual dense search, the query language is not the problem.

4.5 The retriever, the oracle, and the open-weight model

The symmetric monolingual check (English question on the English-only pool against the Indic question on the Indic-only pool, matched queries) is 10.1 points for Hindi and 12.6 for Telugu under dense retrieval and 35.3 and 31.0 under BM25 (all survive), and with the gold passage in the prompt it is 6.3 points for both languages (Hindi survives; Telugu $p = 0.10$): D5 holds. With the passage given, Gemini 2.5 Flash-Lite answers the Hindi question from the English passage 80.5% of the time and from the Hindi passage 72.8%; the model reads English evidence better than Hindi evidence even for a Hindi question. The two stronger Gemini models read as well in Hindi as in English (integration gaps of -2.0 and -2.0 points),

Contrast (percentage points)	Gemini 2.5 Flash-Lite		Gemini 3.1 Flash-Lite		Gemini 3.6 Flash		Qwen3-30B-A3B	
	Hindi	Telugu	Hindi	Telugu	Hindi	Telugu	Hindi	Telugu
Total gap: English minus Indic question (free, both pools)	**+9.4 [+4.5, +14.3]***	+4.7 [-1.6, +11.0]	+3.0 [-2.0, +8.0]	+1.0 [-5.1, +7.1]	+2.0 [-4.0, +10.0]	-6.0 [-14.0, +0.0]	**+9.1 [+3.8, +14.3]***	+2.4 [-4.7, +9.4]
Recovered by dual-language queries	+1.8 [-3.2, +6.8]	-2.5 [-9.2, +4.2]	+1.0 [-4.0, +6.0]	-3.0 [-10.1, +3.0]	+0.0 [-6.0, +6.0]	-2.0 [-8.0, +4.0]	+4.5 [+0.7, +8.4]*	+3.9 [-0.8, +9.4]
Recovered by English-only queries	+4.5 [+0.0, +9.1]	+0.0 [-6.3, +6.3]	+2.0 [-3.0, +7.0]	-1.0 [-6.1, +4.0]	+6.0 [+0.0, +14.0]	-2.0 [-10.0, +6.0]	-1.4 [-6.3, +3.5]	-1.6 [-7.1, +3.9]
Recovered by matched-language queries	+0.7 [-3.5, +4.9]	-7.9 [-14.2, -1.6]*	+1.0 [-3.0, +5.1]	-2.0 [-6.1, +2.0]	–	–	+0.3 [-2.8, +3.5]	-0.8 [-6.3, +4.7]
Evidence-side: both pools minus English-only pool	-0.7 [-5.2, +4.2]	-1.6 [-7.9, +4.7]	-2.0 [-7.0, +3.0]	+1.0 [-3.0, +5.1]	+0.0 [-6.0, +6.0]	+2.0 [-6.0, +10.0]	+5.6 [+1.0, +10.5]*	+2.4 [-3.1, +7.9]
Evidence-side: both pools minus Indic-only pool	+0.3 [-3.5, +4.2]	+3.1 [-2.4, +8.7]	+4.0 [-2.0, +10.1]	+0.0 [-6.1, +6.1]	–	–	+1.4 [-2.1, +4.9]	+3.9 [-2.4, +11.0]
English queries minus matched queries, English-only pool	+6.3 [+1.7, +11.2]*	+2.4 [-3.1, +7.9]	+3.0 [-2.0, +8.0]	+3.0 [-2.0, +8.0]	–	–	+3.8 [-1.0, +8.7]	+2.4 [-3.9, +9.4]
English queries minus matched queries, Indic-only pool	-4.9 [-10.1, +0.7]	+0.8 [-7.1, +8.7]	+0.0 [-5.0, +5.0]	+1.0 [+0.0, +3.0]	–	–	-4.2 [-8.4, +0.0]	-7.1 [-14.2, +0.0]
Symmetric monolingual: en/en minus L/L	**+10.1 [+4.5, +15.7]***	+12.6 [+3.9, +21.3]*	–	–	–	–	**+13.2 [+7.7, +18.8]***	+3.1 [-4.7, +11.0]
Integration gap: oracle passage, English minus Indic question	**+6.3 [+2.8, +9.8]***	+6.3 [+0.0, +12.6]	-2.0 [-6.0, +2.0]	-1.0 [-5.0, +2.0]	-2.0 [-10.0, +4.0]	-2.0 [-6.0, +0.0]	+4.2 [+0.7, +7.7]*	+6.3 [+1.6, +11.0]*
Closed-book gap	+3.1 [-1.0, +7.3]	+3.9 [-2.4, +10.2]	+3.0 [-4.0, +10.0]	-1.0 [-7.0, +5.0]	+2.0 [-4.0, +10.0]	+0.0 [-6.0, +6.0]	**+12.9 [+7.3, +18.5]***	+7.1 [+0.0, +14.2]
Total gap under BM25	**+18.9 [+12.9, +24.8]***	+11.0 [+2.4, +19.7]*	–	–	–	–	**+19.5 [+13.6, +25.4]***	**+14.2 [+6.3, +22.0]***
Symmetric monolingual under BM25	**+35.3 [+28.7, +41.6]***	**+31.0 [+19.8, +42.1]***	–	–	–	–	–	–
Recovered by dual queries under BM25	**+9.3 [+3.9, +14.6]***	+7.6 [-1.7, +16.8]	–	–	–	–	**+13.6 [+8.4, +19.2]***	+9.4 [+2.4, +17.3]*
Indic question: BM25 minus dense (free, both)	**-15.7 [-21.3, -10.1]***	**-21.3 [-30.7, -12.6]***	–	–	–	–	**-16.0 [-21.6, -10.1]***	**-18.9 [-28.3, -10.2]***
Policy × evidence interaction	**+11.2 [+4.2, +18.2]***	+1.6 [-7.9, +11.0]	+3.0 [-4.0, +10.0]	+2.0 [-3.0, +8.0]	–	–	+8.0 [+1.7, +14.3]*	+9.4 [-0.8, +18.9]

Table 2: Paired within-item contrasts (percentage points) with item-bootstrap 95% intervals. *: sign-flip permutation $p < 0.05$. Bold: survives Benjamini–Hochberg over the full family of 134 tests (threshold $p \leq 0.0067$). Positive total gap = the English question is answered more often. Dense retriever unless marked BM25.

so D6 holds for them and fails by 1.3 points for Gemini 2.5 Flash-Lite. Closed-book, the models already know many of these facts (57.8% to 70.0% for Hindi questions, 64.6% to 84.0% for Telugu), with closed-book gaps of 2–4 points; 77% (Hindi) and 76% (Telugu) of correct agent answers on Gemini 2.5 Flash-Lite had a gold chunk in their trajectory. On the pre-specified needs-search subset (97 Hindi items Gemini 2.5 Flash-Lite got wrong closed-book in both languages) the Hindi gap is 12.5 points ($p = 0.02$) and dual queries recover 7.4 ($p = 0.15$). With the gold evidence hidden from the pool the Hindi agent still scores 55.1%, close to its closed-book 57.8%, and abstains in 10% of episodes.

The natively written Telugu questions (TyDi XOR, 120 items, Appendix D) are harder than the translated ones (60.0% against 74.8% on the both-pools cell, different items), but the agent behaves the same way on them: 44% English queries under the free policy against 44–47% on the translated set, and no gain from dual queries. The translation of the questions does not drive the query-language behaviour.

Qwen3-30B-A3B. The open-weight model behaves like the cheaper Gemini with two differences. Its Hindi gap is 9.1 points (survives), its Telugu gap 2.4; dual queries recover 4.5 of the Hindi points ($p = 0.035$, not surviving) and the English-only pool costs it 5.6 points against both pools ($p = 0.03$, not surviving), so a small evidence-side term is visible here where Gemini shows none. It writes only 18% of its queries in English for Hindi questions, searches at all in 92% of episodes, and its closed-book Hindi accuracy is far below its English one (39.4 vs 52.3, a 12.9-point gap that survives), so it depends on retrieval more than Gemini does. Under BM25 the Hindi question costs it 19.5 points and dual queries recover 13.6 (both survive); its integration gap with the English passage given is 4.2 points. Coverage-weighted, a Hindi user of Qwen expects 61.6 with cross-lingual search and 43.3 with keyword search against 74.6 and 69.7 for an English user.

5 Conclusion

We built the instrument that separates what a search agent's queries lose from what the web lacks for Indic users, and measured both on parallel questions, a frozen two-language pool and a logged, then forced, query language. The answer depends on the search engine. With a cross-lingual dense retriever, the Indic user's deficit is not the query and not the web: the served pool and the query policy barely matter, and the residue of 2–9 points is the model's reading of evidence, largest for the cheapest models and absent for the stronger ones. With a keyword retriever the query language decides everything: the Hindi question costs 19–20 points, searching only in Hindi collapses, and searching in English recovers most of the loss, because 71–76% of the facts these users ask about exist only in English Wikipedia. Agents do not discover this on their own; they keep the same mix of query languages whatever the pool returns. For a product team the advice is concrete: find out whether the engine behind the agent is cross-lingual; if it is not, translate the query; if it is, the lever is the model. Everything regenerates from the public snapshot and the scripts for under twelve dollars of metered calls.

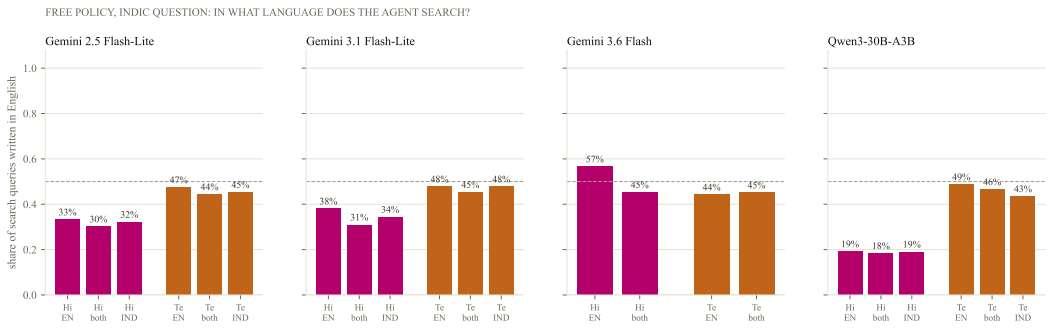


Figure 2: Share of search queries written in Latin script when the question is in Hindi (magenta) or Telugu (orange), free policy, by served pool (EN = English-only, IND = Indic-only). Dashed line: one half.

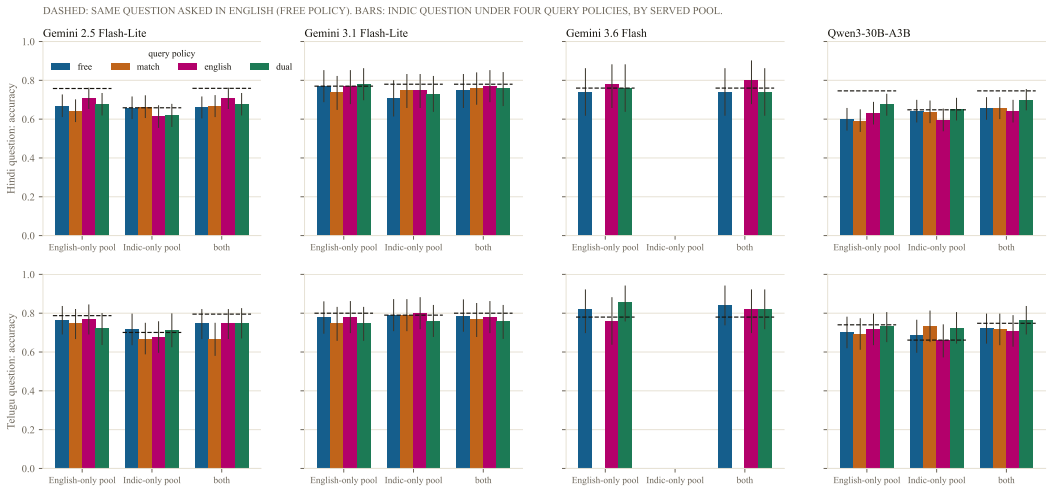


Figure 3: Accuracy of the Indic question under the four query policies and the three served pools, dense retriever, with item-bootstrap 95% intervals; the dashed line is the same question asked in English under the free policy. Top row Hindi, bottom row Telugu.

6 Related work

Agentic search beyond English. Monolingual browsing benchmarks exist for Chinese (Zhou et al., 2025), Korean (Lee et al., 2026a; Jeong, 2026), and thirteen natively authored languages without an Indic one (Aji et al., 2026); BrowseComp-Plus fixed the corpus so that retrievers can be swapped (Chen et al., 2025), and XBCP translated that corpus into twelve languages and measured deep-research agents with oracle retrieval, finding retrieval rather than integration to be the larger loss (Lu et al., 2026). Multilingual GAIA variants place Hindi among the weakest languages at every model size (Hofman et al., 2025; Caciolai et al., 2026) and warn that part of the measured gap is translation artefact (Kim et al., 2026). English-only taxonomies split agent failures into retrieval and utilisation (Liu et al., 2026b; Liang et al., 2026; Liu et al., 2026a); logging what agents send to search engines is rare and English-centred (Amani et al., 2026), and non-English queries receive fewer citations (Nguyen et al., 2026). The MAST shared task evaluates Indic queries over an English corpus (Thakur et al., 2026); MarathiWeb evaluates two web-search models on native Marathi needs (Kale, 2026). None logs or forces an agent's query language for an Indic user, and none varies where the evidence lives.

Language preference in multilingual retrieval. Retrievers and generators favour high-resource and query-language passages (Park and Lee, 2025; Wang et al., 2026; Jang

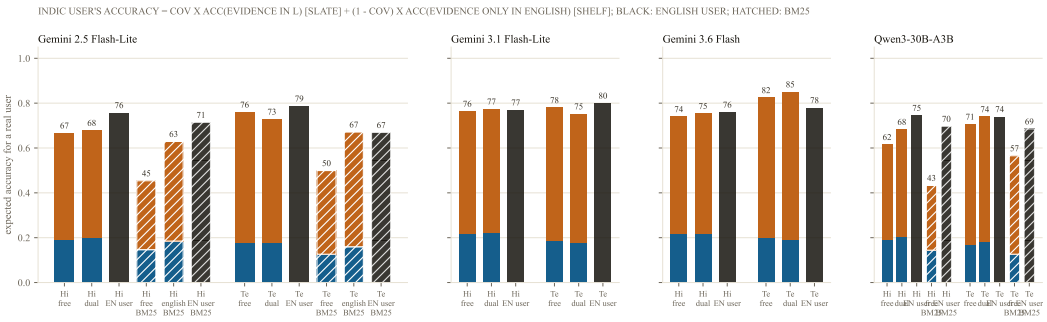


Figure 4: Expected accuracy of a real Indic user from Eq. 1: the slate part is the share of facts the Indic Wikipedia holds times the accuracy with evidence in both languages, the orange part the share only English holds times the accuracy on the English-only pool; black is the English user. Hatched bars use BM25 instead of the dense retriever.

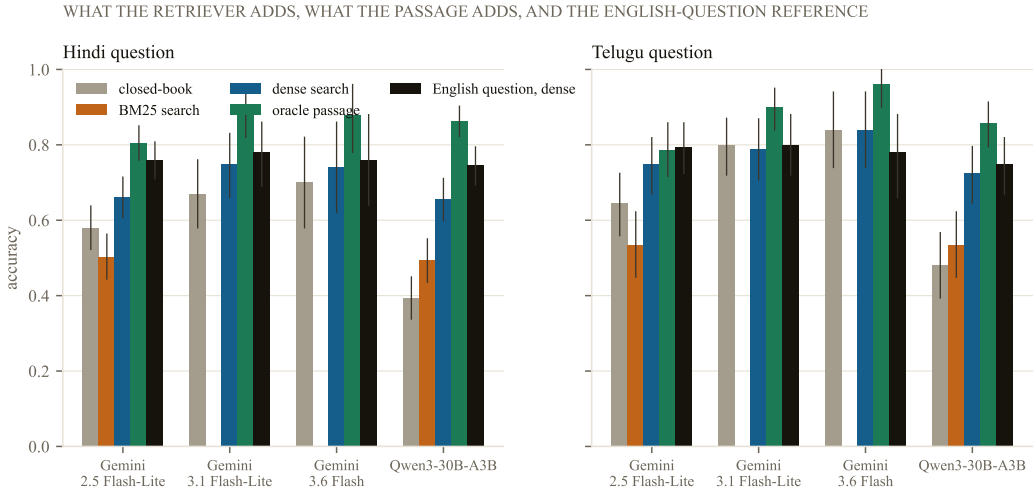


Figure 5: Indic question, free policy, both pools: closed-book, BM25 search, dense search, the gold passage in the prompt, and the English question with dense search, per model.

et al., 2026; Ki et al., 2025), outputs drift toward English when the evidence language mismatches (Li et al., 2025), and reasoning models fail to transfer knowledge across scripts (Bandarkar et al., 2026). DELTA showed that the measured English preference is confounded by where gold evidence exists and corrected for it statistically (Park et al., 2026); we manipulate it experimentally. Pipeline comparisons disagree on whether translating the query helps (Ranaldi et al., 2025; Moon et al., 2025; Gupta et al., 2025), and The Cross-Lingual Cost shows the sign depends on where the documents are (Amiraz et al., 2025); XRAG is the one controlled query-language ablation on a live engine, where native and translated queries were complementary (Liu et al., 2025). CORAL lets an agent choose which-language corpus to search but reports end accuracy only (Lee et al., 2026b). Dense multilingual retrievers retrieve across languages where lexical matching cannot (Goworek et al., 2025; Chen et al., 2024), which is why we treat the retriever as a factor. Up to half of correct cross-lingual QA answers are not attributable to any retrieved passage (Muller et al., 2023), which is why we run a closed-book arm on every item.

Indic benchmarks and corpora. Parallel Indic questions with known evidence come from IndicGenBench (Singh et al., 2024), built on TyDi XOR (Asai et al., 2020; Clark et al., 2020), MLQA (Lewis et al., 2019), and XQuAD (Artetxe et al., 2019); India-centric closed-book sets include IndicQuest (Jain et al., 2026; Rohera et al., 2025), MILU (Verma et al., 2024), and MultiLoKo (Hupkes and Bogoychev, 2025). Indic retrieval resources include

Model	Language	Cov_L	Indic user, free	Indic user, dual	Indic user, english	English user
Gemini 2.5 Flash-Lite	Hindi	29.1%	66.7	67.9	70.7	75.8
under BM25	Hindi	29.1%	45.4	58.4	62.9	71.4
Gemini 2.5 Flash-Lite	Telugu	23.6%	76.0	72.9	76.6	78.7
under BM25	Telugu	23.6%	49.9	55.9	67.1	66.9
Gemini 3.1 Flash-Lite	Hindi	29.1%	76.4	77.4	77.0	77.0
Gemini 3.1 Flash-Lite	Telugu	23.6%	78.0	75.2	78.0	80.0
Gemini 3.6 Flash	Hindi	29.1%	74.0	75.4	78.6	76.0
Gemini 3.6 Flash	Telugu	23.6%	82.5	85.1	77.4	78.0
Qwen3-30B-A3B	Hindi	29.1%	61.6	68.3	63.4	74.6
under BM25	Hindi	29.1%	43.3	59.9	–	69.7
Qwen3-30B-A3B	Telugu	23.6%	70.6	74.0	71.5	74.0
under BM25	Telugu	23.6%	56.6	64.8	–	68.5

Table 3: Expected accuracy for a real user, Eq. 1: Cov_L is the share of the 539 XorQA questions whose Hindi or Telugu Wikipedia article verifiably states the answer; the Indic user's accuracy weights the both-pools cell by Cov_L and the English-only cell by $1-\text{Cov}_L$; the English user is the English question on the English-only pool.

MIRACL (Zhang et al., 2022), IndicIRSuite (Haq et al., 2023), IndicRAGSuite (Prasanjith et al., 2025), and Hindi-BEIR (Acharya et al., 2024); Health-Pariksha evaluates a fixed RAG pipeline on real Indic health queries (Gumma et al., 2024).

Knowledge and coverage gaps. Factual recall in language models depends on language and script (Jiang et al., 2020; Kassner et al., 2021; Chua et al., 2024; Piratla et al., 2025); India-centric questions are answered better in English even by strong models (Rohera et al., 2025), and code-mixing narrows the gap (Mazumder et al., 2026). ECLeKTic and Where's Waldo build questions whose evidence exists in only one Wikipedia edition (Goldman et al., 2025; Ki et al., 2026), the closed-book analogue of our evidence-side manipulation. Wikipedia's coverage asymmetry has been documented since Hecht and Gergle (Hecht and Gergle, 2010) and runs both ways (Wang et al., 2025); our pools keep the local Indic facts that English lacks.

7 Limitations

Wikipedia is not the web. Our pools are complete for Hindi and Telugu but a 1.8% sample of English Wikipedia plus the articles behind the items, so the English pool has fewer near-topic distractors than a real engine would return; the symmetric monolingual cells and the retriever factor bound, but do not remove, this asymmetry. Two languages and one Indic family of scripts each are studied; Bengali, Tamil and the others are not. The XorQA questions are human translations of English questions, so the Indic phrasing is not what a native user would type; the natively written Telugu subset is the check, not a cure. Coverage was measured by string match and two LLM reads of each article with a manual audit of 50 items; the audit found the first read too generous, which is why the second was added, and the remaining uncertainty is reported with the coverage estimate. The closed-model arm is Gemini only because a workspace guardrail on the API account blocked the other closed families during the study; the open-weight arm is one model. Models change; the model identifiers and the snapshot date are recorded. The judge is itself a Gemini model; the string rule decides most items and the audited sample bounds the rest. No live-web arm was run.

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A Verbatim prompts and tool schema

All prompts are reproduced exactly as sent. The policy sentence is appended to the system prompt; `{qlang}` is the name of the question's language and `{max_searches}`=3, `{k}`=5.

Agent system prompt

```
You are a search agent. You answer factual questions by searching a
document collection with the `search` tool, reading the returned
passages, and then giving a short final answer.
Rules:
- You may call `search` at most {max_searches} times. Each call
returns the top {k} passages.
- When you have enough evidence, or when you have used all your
searches, stop calling tools and reply with the final answer only: a
short phrase (a name, date, number, or term), no explanation.
- Answer in the same language as the question.
- If the evidence does not contain the answer, you may still answer
from your own knowledge, but prefer the evidence.{policy}
```

Policy sentence: free

```
(none)
```

Policy sentence: match

```
- Write every search query in {qlang}, the language of the question.
```

Policy sentence: english

```
- Write every search query in English, even though the question is
in {qlang}. Translate the information need into an English query.
```

Policy sentence: dual

```
- At every step call `search` twice: once with a query in {qlang}
(the language of the question) and once with the same information
need as an English query. Both calls count as one step.
```

Tool schema (JSON)

```
{
"type": "function",
"function": {
"name": "search",
"description": "Search the document collection. Returns the most
relevant passages with their titles.",
"parameters": {
"type": "object",
"properties": {
"query": {
"type": "string",
"description": "The search query."
}
},
"required": [
"query"
]
```

```
}
}
}
```

Empty-result message

```
No passages found for this query.
```

Closed-book system prompt

```
You answer factual questions from your own knowledge. Reply with the
final answer only: a short phrase (a name, date, number, or term),
no explanation. Answer in the same language as the question.
```

Oracle system prompt

```
You answer a factual question using the passage provided. Reply with
the final answer only: a short phrase (a name, date, number, or
term), no explanation. Answer in the same language as the question.
```

Oracle user message

```
Passage:
"""
{passage}
"""
Question: {question}
```

Judge system prompt

```
You are a strict grader of short answers to factual questions.
Decide whether the candidate answer expresses the same fact as any
of the gold answers. Accept equivalent forms: a transliteration or
translation of the gold answer into another language or script, a
date written differently, a number with or without units, a name
with or without honorifics, a more specific answer that contains the
gold answer. Reject answers that name a different entity, date, or
number, that hedge between several candidates. If the candidate
declines to answer, says the information is unavailable, or asks a
question back, that is ABSTAIN.
```

Judge user message

```
Question (English): {q_en}
Question (original): {q_L}
Gold answers: {gold}
Candidate answer: {cand}
Reply with exactly one word: CORRECT, WRONG, or ABSTAIN (the
candidate says it does not know, cannot find the answer, or asks for
clarification instead of answering).
```

B Coverage measurement and audit

For each of the 539 XorQA-In test questions per language we resolved the English source article to its Hindi or Telugu counterpart through Wikidata sitelinks (916 of 918 distinct titles resolved; two had no Wikidata item) and looked the counterpart up in the 20261001 snapshot. An article exists for 364 Hindi and 283 Telugu items. Coverage was then decided in three steps. (1) *String match*: the Indic gold answer, or the English gold answer, occurs in the article after NFC normalisation and punctuation stripping; 119 Hindi and 94 Telugu

items (the 237 Telugu items whose released translated answer is the function word "అని" were matched on the English answer only). (2) *Two independent LLM reads*: gemini-2.5-flash-lite and then gemini-3.6-flash were each given the question in both languages, the gold answers and the full article (up to 14,000 characters) and asked to quote verbatim the sentence that states the answer or reply NONE; a read counts only if the quoted sentence is found in the article. The first read accepted 205 Hindi and 179 Telugu items; the second accepted 158 and 129; both agree on 157 and 127, which are the verified-coverage counts (29.1% and 23.6%). (3) *Manual audit*: before the second read, we drew 25 items per language stratified over first-read outcomes and labelled them by reading the quoted sentence against the question. Of the 30 first-read acceptances, 21 were correct, 7 were uncertain (the quote did not visibly contain the answer) and 2 were wrong (the Hindi article gives 13 South American countries where the gold says twelve; the Telugu Gandhi article never names Subhas Chandra Bose). The second read rejected 1 of the 2 wrong, 4 of the 7 uncertain and 8 of the 21 correct acceptances, so the final coverage figure is conservative: it under-counts rather than over-counts. The 20 first-read rejections in the audit were not labelled by hand; the second read accepted 3 of them with a verbatim quote, but an item needs both reads to count, so they stay outside the controlled grid and the coverage figure. The audit file with every label is shipped as `results/coverage_audit.json`.

C Judge audit

Of the judged agent answers, the string rule decided 41% (exact or contained match to a gold alias in either script after digit normalisation); the rest went to the LLM judge (gemini-2.5-flash, not in the grid), which returned CORRECT, WRONG or ABSTAIN. We drew 40 judged answers (20 per language: 10 string-correct, 5 LLM-correct, 5 LLM-wrong or abstain) and labelled them by hand against the question and gold aliases. All 40 verdicts agreed with the manual label. Cases worth noting: a Korean-script answer accepted for the romanised gold "mumu"; "labour movement" accepted for "trade union movement"; an answer naming one of three states in the gold marked WRONG, which we consider defensible; and seven abstentions ("the search results do not contain…") marked ABSTAIN and scored as wrong. The sample is shipped as `results/audit_sample.json`.

D Per-cell tables

E Cost

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	287	57.8	[52.3, 63.8]	0%	0%
Hindi	dual	Indic-only	281	61.9	[56.2, 67.6]	50%	79%
Hindi	dual	Indic-only [bm25]	280	34.3	[28.6, 40.0]	52%	20%
Hindi	dual	both	280	67.9	[62.1, 73.2]	50%	87%
Hindi	dual	both [bm25]	281	58.7	[52.7, 64.4]	51%	67%
Hindi	dual	English-only	280	67.9	[62.1, 73.2]	50%	83%
Hindi	dual	English-only [bm25]	280	58.2	[52.5, 63.9]	51%	61%
Hindi	english	Indic-only	287	61.3	[55.7, 66.9]	100%	68%
Hindi	english	Indic-only [bm25]	285	32.3	[27.0, 37.9]	100%	8%
Hindi	english	both	287	70.7	[65.5, 76.0]	100%	80%
Hindi	english	both [bm25]	287	63.4	[57.8, 69.0]	100%	63%
Hindi	english	English-only	287	70.7	[65.5, 76.0]	100%	80%
Hindi	english	English-only [bm25]	287	62.7	[57.1, 68.3]	100%	59%
Hindi	free	Indic-only	287	65.9	[60.3, 71.4]	32%	74%
Hindi	free	Indic-only [bm25]	287	40.1	[34.5, 45.6]	38%	17%
Hindi	free	both	287	66.2	[60.6, 71.4]	30%	79%
Hindi	free	both [bm25]	286	50.3	[44.4, 56.3]	28%	40%
Hindi	free	English-only	287	66.9	[61.3, 72.5]	33%	72%
Hindi	free	English-only [bm25]	286	43.4	[37.4, 49.3]	35%	38%
Hindi	free	(gold hidden)	49	55.1	[40.8, 69.4]	27%	0%
Hindi	match	Indic-only	286	66.4	[60.8, 72.0]	1%	74%
Hindi	match	Indic-only [bm25]	286	35.3	[29.7, 40.9]	1%	16%
Hindi	match	both	287	66.9	[61.3, 72.1]	1%	78%
Hindi	match	both [bm25]	287	43.6	[38.0, 49.5]	2%	23%
Hindi	match	English-only	286	64.3	[58.7, 69.9]	1%	66%
Hindi	match	English-only [bm25]	286	19.6	[15.0, 24.1]	2%	3%
Hindi	oracle L	(gold passage)	287	72.8	[67.6, 77.7]	0%	0%
Hindi	oracle en	(gold passage)	287	80.5	[76.0, 85.0]	0%	0%
English	closed	(no tool)	287	61.0	[55.4, 66.6]	0%	0%
English	dual	Indic-only	287	64.5	[58.9, 70.0]	97%	72%
English	dual	Indic-only [bm25]	287	25.8	[20.9, 31.0]	97%	8%
English	free	Indic-only	287	65.9	[60.3, 71.4]	100%	70%
English	free	Indic-only [bm25]	286	32.2	[26.9, 37.4]	100%	8%
English	free	both	286	75.9	[71.0, 80.8]	100%	83%
English	free	both [bm25]	287	69.0	[63.4, 74.2]	100%	65%
English	free	English-only	285	75.8	[70.5, 80.7]	100%	82%
English	free	English-only [bm25]	287	71.4	[66.2, 76.3]	100%	65%
English	match	Indic-only	287	64.8	[59.2, 70.4]	100%	70%
English	match	Indic-only [bm25]	287	38.3	[32.8, 43.9]	100%	8%
English	match	both	287	74.9	[69.7, 79.8]	100%	82%
English	match	both [bm25]	287	71.4	[65.9, 76.7]	100%	65%
English	match	English-only	287	76.3	[71.4, 81.2]	100%	82%
English	match	English-only [bm25]	287	70.7	[65.5, 76.0]	100%	64%
English	oracle en	(gold passage)	287	86.8	[82.6, 90.6]	0%	0%

Table 4: All cells, Gemini 2.5 Flash-Lite, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	127	64.6	[55.9, 72.4]	0%	0%
Telugu	dual	Indic-only	118	71.2	[62.7, 79.7]	50%	81%
Telugu	dual	Indic-only [bm25]	118	44.9	[36.4, 54.2]	51%	13%
Telugu	dual	both	119	74.8	[67.2, 82.4]	50%	89%
Telugu	dual	both [bm25]	119	61.3	[52.1, 69.7]	50%	60%
Telugu	dual	English-only	119	72.3	[63.9, 79.8]	50%	89%
Telugu	dual	English-only [bm25]	118	54.2	[45.8, 63.6]	51%	55%
Telugu	english	Indic-only	127	67.7	[59.8, 75.6]	100%	65%
Telugu	english	Indic-only [bm25]	127	44.1	[35.4, 52.8]	100%	5%
Telugu	english	both	127	74.8	[66.9, 81.9]	100%	87%
Telugu	english	both [bm25]	127	67.7	[59.1, 75.6]	100%	62%
Telugu	english	English-only	127	77.2	[69.3, 84.3]	100%	87%
Telugu	english	English-only [bm25]	127	66.9	[58.3, 74.8]	100%	61%
Telugu	free	Indic-only	127	71.7	[63.8, 79.5]	45%	72%
Telugu	free	Indic-only [bm25]	127	43.3	[34.6, 52.0]	51%	11%
Telugu	free	both	127	74.8	[66.9, 81.9]	44%	81%
Telugu	free	both [bm25]	127	53.5	[44.9, 62.2]	43%	40%
Telugu	free	English-only	127	76.4	[69.3, 83.5]	47%	86%
Telugu	free	English-only [bm25]	127	48.8	[40.2, 57.5]	46%	43%
Telugu	free	(gold hidden)	50	72.0	[60.0, 84.0]	40%	0%
Telugu	match	Indic-only	127	66.9	[59.1, 74.8]	0%	74%
Telugu	match	Indic-only [bm25]	127	37.8	[29.1, 46.5]	2%	11%
Telugu	match	both	127	66.9	[58.3, 74.8]	0%	76%
Telugu	match	both [bm25]	127	35.4	[26.8, 44.1]	2%	14%
Telugu	match	English-only	127	74.8	[66.9, 81.9]	3%	80%
Telugu	match	English-only [bm25]	127	15.0	[9.4, 21.3]	2%	4%
Telugu	oracle L	(gold passage)	127	70.9	[63.0, 78.7]	0%	0%
Telugu	oracle en	(gold passage)	127	78.7	[71.7, 85.8]	0%	0%
English	closed	(no tool)	127	68.5	[60.6, 76.4]	0%	0%
English	dual	Indic-only	127	64.6	[55.9, 72.4]	96%	66%
English	dual	Indic-only [bm25]	127	32.3	[24.4, 40.9]	96%	9%
English	free	Indic-only	127	70.1	[61.4, 78.0]	100%	65%
English	free	Indic-only [bm25]	126	43.7	[34.9, 52.4]	100%	7%
English	free	both	127	79.5	[72.4, 85.8]	100%	88%
English	free	both [bm25]	127	64.6	[55.9, 73.2]	100%	58%
English	free	English-only	127	78.7	[71.7, 85.8]	100%	88%
English	free	English-only [bm25]	127	66.9	[58.3, 74.8]	100%	59%
English	match	Indic-only	127	70.1	[62.2, 78.0]	100%	60%
English	match	Indic-only [bm25]	126	46.0	[37.3, 54.8]	100%	6%
English	match	both	127	80.3	[73.2, 86.6]	100%	87%
English	match	both [bm25]	127	68.5	[60.6, 76.4]	100%	61%
English	match	English-only	127	79.5	[72.4, 86.6]	100%	87%
English	match	English-only [bm25]	126	69.0	[61.1, 77.0]	100%	60%
English	oracle en	(gold passage)	127	85.0	[78.7, 90.6]	0%	0%

Table 5: All cells, Gemini 2.5 Flash-Lite, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	100	67.0	[58.0, 76.0]	0%	0%
Hindi	dual	Indic-only	100	73.0	[64.0, 82.0]	51%	86%
Hindi	dual	both	100	76.0	[67.0, 84.0]	50%	90%
Hindi	dual	English-only	100	78.0	[70.0, 86.0]	50%	90%
Hindi	english	Indic-only	100	75.0	[66.0, 83.0]	100%	75%
Hindi	english	both	100	77.0	[69.0, 85.0]	100%	86%
Hindi	english	English-only	100	77.0	[68.0, 85.0]	100%	86%
Hindi	free	Indic-only	99	70.7	[61.6, 79.8]	34%	79%
Hindi	free	both	100	75.0	[66.0, 83.0]	31%	83%
Hindi	free	English-only	100	77.0	[69.0, 85.0]	38%	83%
Hindi	match	Indic-only	100	75.0	[66.0, 83.0]	5%	79%
Hindi	match	both	99	75.8	[67.7, 83.8]	7%	82%
Hindi	match	English-only	100	74.0	[65.0, 82.0]	11%	81%
Hindi	oracle L	(gold passage)	100	84.0	[77.0, 91.0]	0%	0%
Hindi	oracle en	(gold passage)	100	89.0	[82.0, 95.0]	0%	0%
English	closed	(no tool)	100	70.0	[61.0, 79.0]	0%	0%
English	free	Indic-only	100	78.0	[69.0, 86.0]	100%	76%
English	free	both	100	78.0	[69.0, 86.0]	100%	86%
English	free	English-only	100	77.0	[68.0, 85.0]	97%	84%
English	oracle en	(gold passage)	100	87.0	[80.0, 93.0]	0%	0%

Table 6: All cells, Gemini 3.1 Flash-Lite, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	100	80.0	[72.0, 87.0]	0%	0%
Telugu	dual	Indic-only	100	76.0	[68.0, 84.0]	51%	78%
Telugu	dual	both	100	76.0	[67.0, 84.0]	50%	92%
Telugu	dual	English-only	100	75.0	[66.0, 83.0]	50%	90%
Telugu	english	Indic-only	100	80.0	[72.0, 88.0]	100%	67%
Telugu	english	both	100	78.0	[70.0, 86.0]	100%	87%
Telugu	english	English-only	100	78.0	[70.0, 86.0]	100%	87%
Telugu	free	Indic-only	100	79.0	[71.0, 87.0]	48%	76%
Telugu	free	both	99	78.8	[70.7, 86.9]	45%	85%
Telugu	free	English-only	99	77.8	[69.7, 85.9]	48%	84%
Telugu	match	Indic-only	100	79.0	[71.0, 87.0]	4%	74%
Telugu	match	both	100	77.0	[68.0, 85.0]	4%	76%
Telugu	match	English-only	100	75.0	[66.0, 83.0]	6%	83%
Telugu	oracle L	(gold passage)	100	86.0	[79.0, 92.0]	0%	0%
Telugu	oracle en	(gold passage)	100	90.0	[84.0, 95.0]	0%	0%
English	closed	(no tool)	100	79.0	[71.0, 87.0]	0%	0%
English	free	Indic-only	100	79.0	[71.0, 87.0]	100%	68%
English	free	both	100	80.0	[72.0, 88.0]	100%	91%
English	free	English-only	100	80.0	[72.0, 87.0]	100%	91%
English	oracle en	(gold passage)	100	89.0	[82.0, 95.0]	0%	0%

Table 7: All cells, Gemini 3.1 Flash-Lite, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	50	70.0	[58.0, 82.0]	0%	0%
Hindi	dual	both	50	74.0	[62.0, 86.0]	50%	88%
Hindi	dual	English-only	50	76.0	[64.0, 88.0]	53%	94%
Hindi	english	both	50	80.0	[68.0, 90.0]	98%	90%
Hindi	english	English-only	50	78.0	[66.0, 88.0]	100%	90%
Hindi	free	both	50	74.0	[62.0, 86.0]	45%	80%
Hindi	free	English-only	50	74.0	[62.0, 86.0]	57%	84%
Hindi	oracle L	(gold passage)	50	78.0	[66.0, 88.0]	0%	0%
Hindi	oracle en	(gold passage)	50	88.0	[78.0, 96.0]	0%	0%
English	closed	(no tool)	50	72.0	[60.0, 84.0]	0%	0%
English	free	both	50	76.0	[64.0, 88.0]	100%	90%
English	free	English-only	50	76.0	[64.0, 88.0]	100%	84%
English	oracle en	(gold passage)	50	86.0	[76.0, 94.0]	0%	0%

Table 8: All cells, Gemini 3.6 Flash, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	50	84.0	[74.0, 94.0]	0%	0%
Telugu	dual	both	50	82.0	[72.0, 92.0]	50%	94%
Telugu	dual	English-only	50	86.0	[76.0, 94.0]	51%	94%
Telugu	english	both	50	82.0	[70.0, 92.0]	98%	94%
Telugu	english	English-only	50	76.0	[64.0, 88.0]	100%	88%
Telugu	free	both	50	84.0	[74.0, 94.0]	45%	80%
Telugu	free	English-only	50	82.0	[70.0, 92.0]	44%	92%
Telugu	oracle L	(gold passage)	50	86.0	[76.0, 94.0]	0%	0%
Telugu	oracle en	(gold passage)	50	96.0	[90.0, 100.0]	0%	0%
English	closed	(no tool)	50	84.0	[74.0, 94.0]	0%	0%
English	free	both	50	78.0	[66.0, 88.0]	100%	84%
English	free	English-only	50	78.0	[66.0, 88.0]	100%	88%
English	oracle en	(gold passage)	50	94.0	[86.0, 100.0]	0%	0%

Table 9: All cells, Gemini 3.6 Flash, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	287	39.4	[33.8, 44.9]	0%	0%
Hindi	dual	Indic-only	287	65.2	[59.6, 70.7]	50%	79%
Hindi	dual	both	287	70.0	[64.8, 75.3]	50%	87%
Hindi	dual	both [bm25]	287	63.1	[57.5, 68.6]	50%	62%
Hindi	dual	English-only	287	67.6	[62.0, 72.8]	50%	85%
Hindi	dual	English-only [bm25]	287	58.5	[53.0, 64.1]	50%	57%
Hindi	english	Indic-only	287	59.6	[54.0, 65.2]	100%	64%
Hindi	english	both	287	64.1	[58.5, 69.7]	100%	79%
Hindi	english	English-only	287	63.1	[57.5, 68.6]	100%	78%
Hindi	free	Indic-only	287	64.1	[58.5, 69.7]	19%	65%
Hindi	free	both	287	65.5	[59.9, 71.1]	18%	69%
Hindi	free	both [bm25]	287	49.5	[43.6, 55.1]	19%	23%
Hindi	free	English-only	287	59.9	[54.4, 65.5]	19%	59%
Hindi	free	English-only [bm25]	287	40.8	[35.2, 46.3]	19%	11%
Hindi	match	Indic-only	287	63.8	[58.2, 69.3]	1%	72%
Hindi	match	both	287	65.9	[60.3, 71.1]	0%	74%
Hindi	match	both [bm25]	287	43.2	[37.6, 48.8]	0%	15%
Hindi	match	English-only	287	59.2	[53.7, 64.8]	0%	65%
Hindi	match	English-only [bm25]	287	37.3	[31.7, 42.9]	0%	0%
Hindi	oracle L	(gold passage)	287	81.9	[77.4, 86.1]	0%	0%
Hindi	oracle en	(gold passage)	287	86.4	[82.2, 90.2]	0%	0%
English	closed	(no tool)	287	52.3	[46.7, 58.2]	0%	0%
English	dual	Indic-only	287	70.7	[65.5, 76.0]	100%	74%
English	free	Indic-only	287	64.8	[59.2, 70.4]	100%	55%
English	free	both	287	74.6	[69.3, 79.4]	100%	71%
English	free	both [bm25]	287	69.0	[63.8, 74.2]	100%	53%
English	free	English-only	287	74.6	[69.3, 79.4]	100%	71%
English	free	English-only [bm25]	287	69.7	[64.5, 74.9]	100%	52%
English	match	Indic-only	287	67.9	[62.4, 73.2]	100%	64%
English	match	both	287	75.6	[70.4, 80.5]	100%	80%
English	match	English-only	287	77.0	[72.1, 81.5]	100%	80%
English	oracle en	(gold passage)	287	90.6	[87.1, 93.7]	0%	0%

Table 10: All cells, Qwen3-30B-A3B, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	127	48.0	[39.4, 56.7]	0%	0%
Telugu	dual	Indic-only	127	72.4	[64.6, 80.3]	50%	80%
Telugu	dual	both	127	76.4	[69.3, 83.5]	50%	93%
Telugu	dual	both [bm25]	127	63.0	[54.3, 71.7]	50%	62%
Telugu	dual	English-only	127	73.2	[65.4, 80.3]	50%	94%
Telugu	dual	English-only [bm25]	127	65.4	[56.7, 73.2]	50%	61%
Telugu	english	Indic-only	127	66.1	[57.5, 74.0]	100%	61%
Telugu	english	both	127	70.9	[63.0, 78.7]	100%	87%
Telugu	english	English-only	127	71.7	[63.8, 79.5]	100%	87%
Telugu	free	Indic-only	127	68.5	[59.8, 76.4]	43%	61%
Telugu	free	both	127	72.4	[64.6, 79.5]	46%	70%
Telugu	free	both [bm25]	127	53.5	[44.9, 62.2]	47%	28%
Telugu	free	English-only	127	70.1	[62.2, 78.0]	49%	71%
Telugu	free	English-only [bm25]	127	57.5	[48.8, 66.1]	46%	24%
Telugu	match	Indic-only	127	73.2	[65.4, 81.1]	1%	74%
Telugu	match	both	127	71.7	[63.8, 79.5]	2%	78%
Telugu	match	both [bm25]	126	41.3	[32.5, 50.0]	1%	9%
Telugu	match	English-only	127	69.3	[61.4, 77.2]	1%	77%
Telugu	match	English-only [bm25]	127	37.8	[29.9, 46.5]	1%	1%
Telugu	oracle L	(gold passage)	127	85.0	[78.7, 90.6]	0%	0%
Telugu	oracle en	(gold passage)	127	85.8	[79.5, 91.3]	0%	0%
English	closed	(no tool)	127	55.1	[46.5, 63.8]	0%	0%
English	dual	Indic-only	127	71.7	[63.8, 79.5]	100%	71%
English	free	Indic-only	127	66.1	[58.3, 74.0]	100%	47%
English	free	both	127	74.8	[66.9, 81.9]	100%	69%
English	free	both [bm25]	127	67.7	[59.8, 75.6]	100%	42%
English	free	English-only	127	74.0	[66.1, 81.1]	100%	70%
English	free	English-only [bm25]	127	68.5	[60.6, 76.4]	100%	43%
English	match	Indic-only	127	69.3	[61.4, 77.2]	100%	57%
English	match	both	127	76.4	[69.3, 83.5]	100%	79%
English	match	English-only	127	76.4	[68.5, 83.5]	100%	83%
English	oracle en	(gold passage)	127	92.1	[87.4, 96.9]	0%	0%

Table 11: All cells, Qwen3-30B-A3B, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Stage	USD
core	9.21
judge	0.79
coverage2	0.63
oracle	0.22
native_te	0.22
coverage	0.19
closed	0.08
noise	0.08
none	0.03
pilot	0.03
filter	0.01
Total metered through OpenRouter (88212 calls)	11.49

Table 12: Metered API spend (exact per-call `usage.cost`). The open-weight arm ran on a rented A100 (about 2.5 GPU-hours) and the bge-m3 embeddings took about 0.5 GPU-hour; both outside the metered total.